

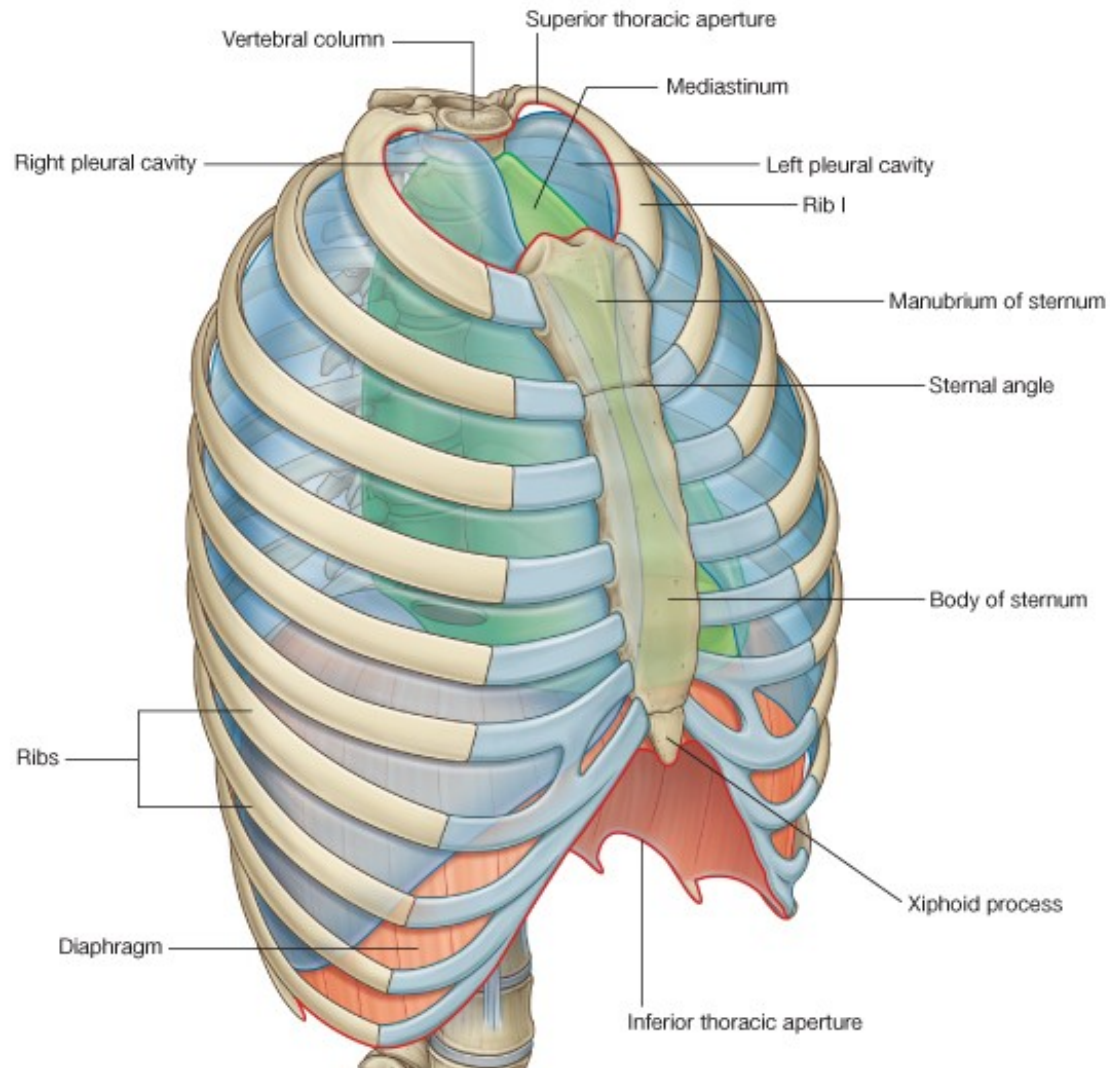
THORAX

Conceptual overview:

GENERAL DESCRIPTION

- The **thorax** is an irregularly shaped cylinder with a narrow opening (superior thoracic aperture) superiorly and a relatively large opening (inferior thoracic aperture) inferiorly.
- The superior thoracic aperture is open, allowing continuity with the neck; the inferior thoracic aperture is closed by the diaphragm.

Thoracic wall and cavity



Thoracic wall and cavity

- The musculoskeletal wall of the thorax is flexible and consists of segmentally arranged vertebrae, ribs, muscles, and the sternum.
- The **thoracic cavity** enclosed by the thoracic wall and the diaphragm is subdivided into three major compartments: a left and a right pleural cavity, each surrounding a lung; mediastinum.

Thoracic wall and cavity

- The mediastinum is a thick, flexible soft tissue partition oriented longitudinally in a median sagittal position.
- It contains the heart, esophagus, trachea, major nerves, and major systemic blood vessels.

Thoracic wall and cavity

- Another important feature of the pleural cavities is that they extend above the level of rib I.
- The apex of each lung actually extends into the root of the neck.
- As a consequence, abnormal events in the root of the neck can involve adjacent pleura and lung, and events in the adjacent pleura and lung can involve the root of the neck.

FUNCTIONS : Breathing

- One of the most important functions of the thorax is breathing.
- The thorax not only contains the lungs but also provides the machinery necessary-the diaphragm, thoracic wall, and the ribs-for effectively moving air into and out of the lungs.
- Up and down movements of the diaphragm and changes in the lateral and anterior dimensions of the thoracic wall, caused by movements of the ribs, alter the volume of the thoracic cavity and are key elements in breathing.

Protection of vital organs

- The thorax houses and protects the heart, lungs, and great vessels.
- Because of the domed shape of the diaphragm, the thoracic wall also offers protection to some important abdominal viscera.
- Much of the liver lies under the right dome of the diaphragm, and the stomach and spleen lie under the left.
- The posterior aspects of the superior poles of the kidneys lie on the diaphragm and are anterior to rib XII, on the right, and to ribs XI and XII, on the left.

Conduit

- The mediastinum acts as a conduit for structures that pass completely through the thorax from one body region to another and for structures that connect organs in the thorax to other body regions.
- The esophagus, vagus nerves, and thoracic duct pass through the mediastinum as they course between the abdomen and neck.
- The phrenic nerves, which originate in the neck, also pass through the mediastinum to penetrate and supply the diaphragm.

Conduit

- Other structures such as the trachea, thoracic aorta, and superior vena cava course within the mediastinum en route to and from major visceral organs in the thorax.

COMPONENT PARTS

Thoracic wall

- The **thoracic wall** consists of skeletal elements and muscles.
 - posteriorly, it is made up of twelve thoracic vertebrae and their intervening intervertebral discs;
 - laterally, the wall is formed by **ribs** (twelve on each side) and three layers of flat muscles, which span the intercostal spaces between adjacent ribs, move the ribs and provide support for the intercostal spaces;
 - anteriorly, the **sternum**, which consists of the manubrium of sternum, body of sternum, and xiphoid process

COMPONENT PARTS

- *The manubrium of sternum, angled posteriorly on the body of sternum at the manubriosternal joint, forms the sternal angle, which is a major surface landmark used by clinicians in performing physical examinations of the thorax.*
- The anterior (distal) end of each rib is composed of costal cartilage, which contributes to the mobility and elasticity of the wall.

COMPONENT PARTS

- All ribs articulate with thoracic vertebrae posteriorly.
- Most ribs (from rib II to IX) have three articulations with the vertebral column. The head of each rib articulates with the body of its own vertebra and with the body of the vertebra above.
- As these ribs curve posteriorly, each also articulates with the transverse process of its vertebra.

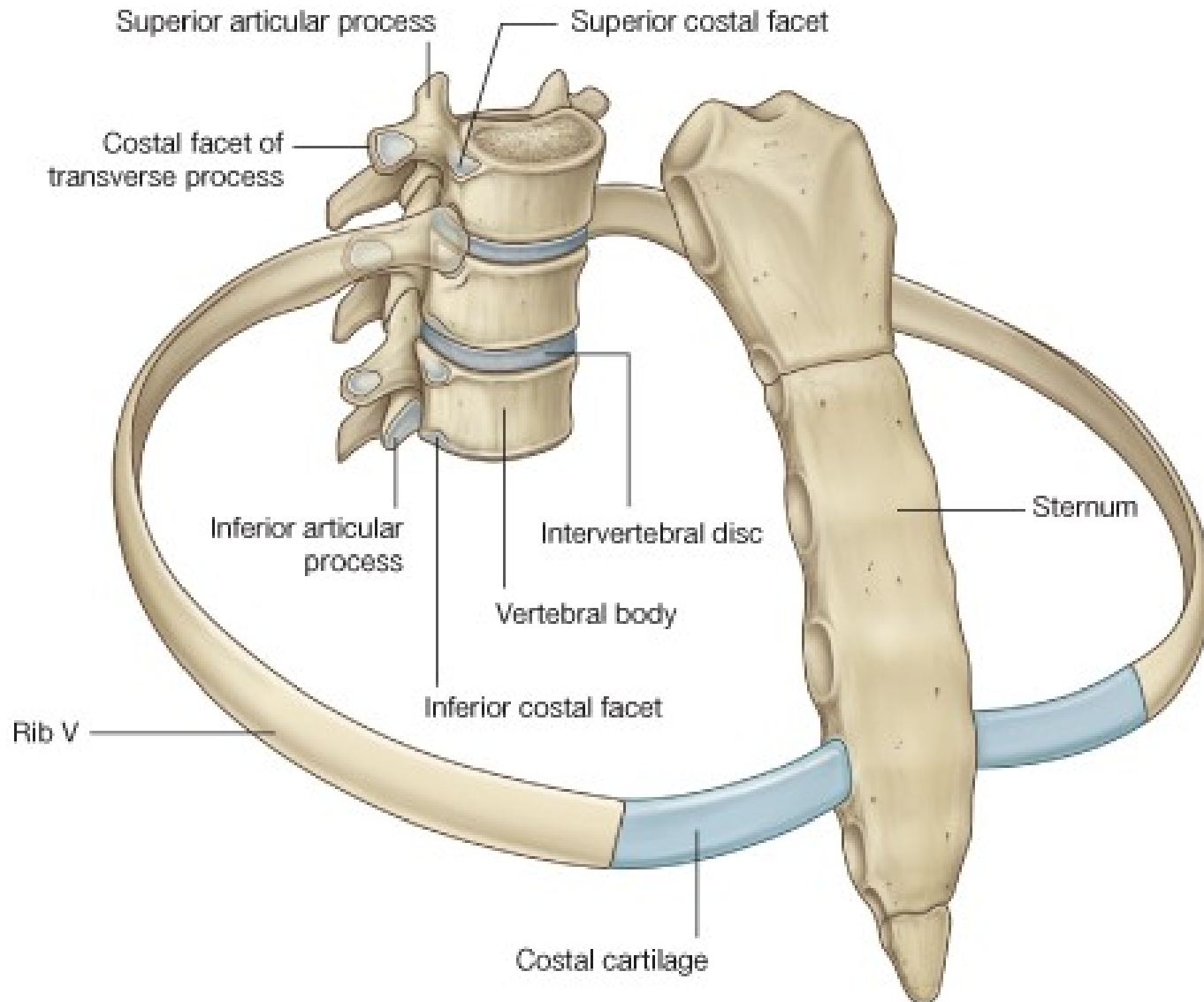
COMPONENT PARTS

- Anteriorly, the costal cartilages of ribs I to VII articulate with the sternum.
- The costal cartilages of ribs VIII to X articulate with the inferior margins of the costal cartilages above them.
- Ribs XI and XII are called floating ribs because they do not articulate with other ribs, costal cartilages, or with sternum.
- Their costal cartilages are small, only covering their tips.

COMPONENT PARTS

- The skeletal framework of the thoracic wall provides extensive attachment sites for muscles of the neck, abdomen, back, and upper limbs.
- A number of these muscles attach to ribs and function as accessory respiratory muscles; some of them also stabilize the position of the first and last ribs.

Joints between ribs and vertebrae



Superior thoracic aperture

- Completely surrounded by skeletal elements, the **superior thoracic aperture** consists of the body of vertebra T1 posteriorly, the medial margin of rib I on each side, and the manubrium anteriorly.
- The superior margin of the manubrium is in approximately the same horizontal plane as the intervertebral disc between vertebrae T2 and T3.
- The first ribs slope inferiorly from their posterior articulation with vertebra T1 to their anterior attachment to the manubrium.
- Consequently, the plane of the superior thoracic aperture is at an oblique angle, facing somewhat anteriorly.

Superior thoracic aperture

- At the superior thoracic aperture, the superior aspects of the pleural cavities, which surround the lungs, lie on either side of the entrance to the mediastinum.
- Structures that pass between the upper limb and thorax pass over rib I and the superior part of the pleural cavity as they enter and leave the mediastinum.
- Structures that pass between the neck and head and the thorax pass more vertically through the superior thoracic aperture.

Inferior thoracic aperture

- The **inferior thoracic aperture** is large and expandable.
- Bone, cartilage, and ligaments form its margin.
- The inferior thoracic aperture is closed by the diaphragm, and structures passing between the abdomen and thorax pierce or pass posteriorly to the diaphragm.

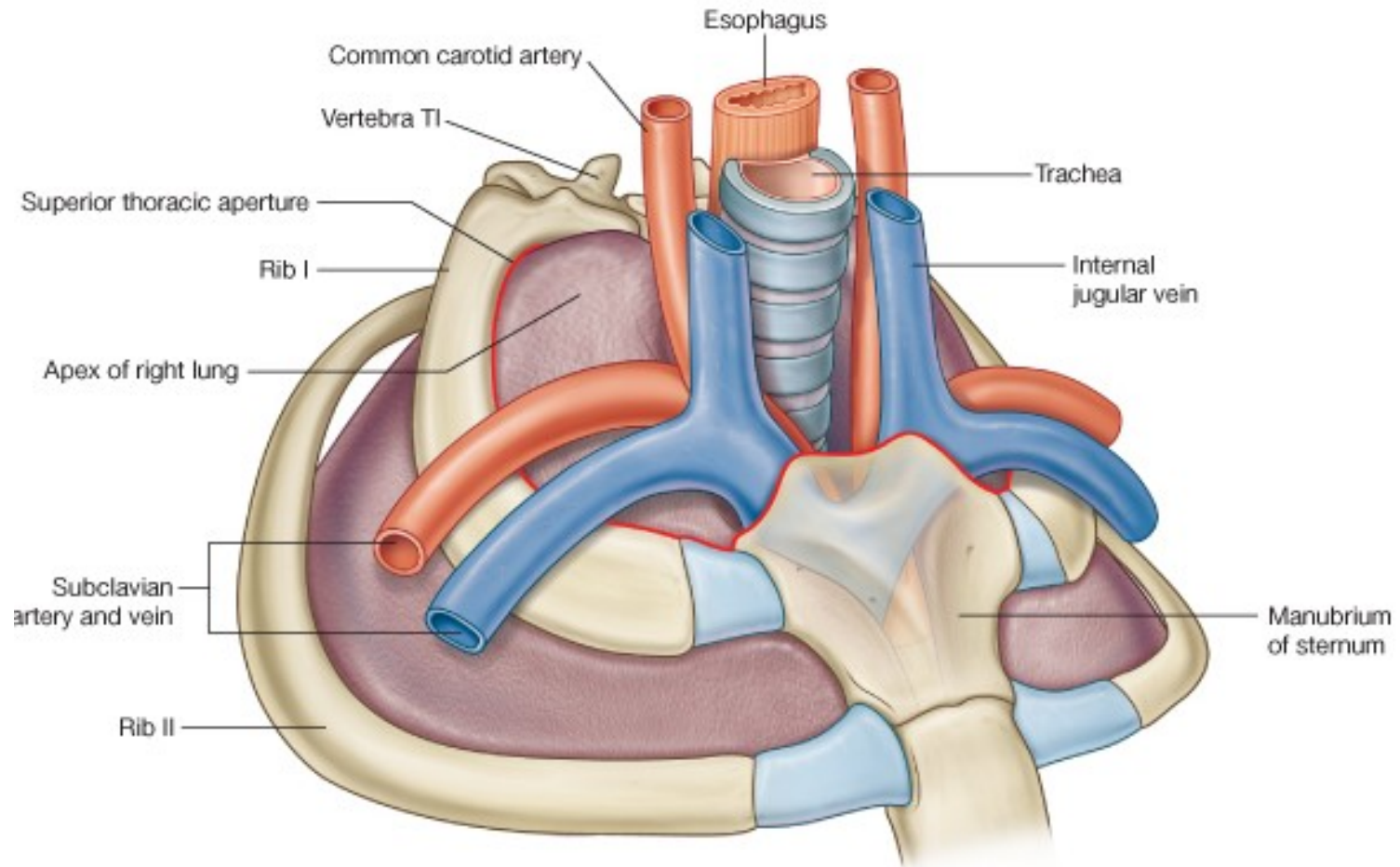
Skeletal elements of the inferior thoracic aperture are:

- the body of vertebra TXII posteriorly;
- rib XII and the distal end of rib XI posterolaterally;
- the distal cartilaginous ends of ribs VII to X, which unite to form the costal margin anterolaterally;
- the xiphoid process anteriorly.

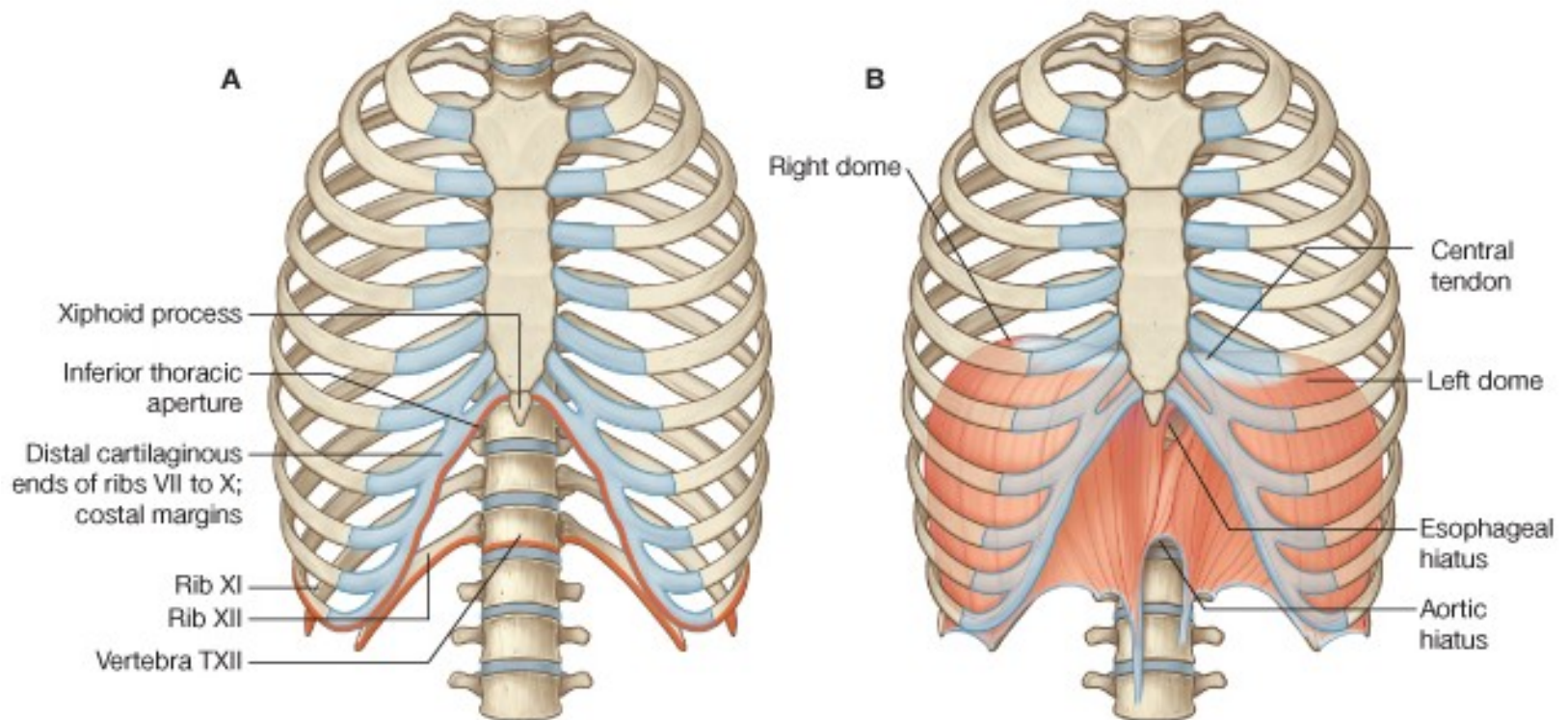
Skeletal elements of the inferior thoracic aperture are:

- The joint between the costal margin and sternum lies roughly in the same horizontal plane as the intervertebral disc between vertebrae TIX and TX.
- In other words, the posterior margin of the inferior thoracic aperture is inferior to the anterior margin.

When viewed anteriorly, the inferior thoracic aperture is tilted superiorly.



Superior thoracic aperture



Diaphragm

- The musculotendinous **diaphragm** seals the inferior thoracic aperture.
- Generally, muscle fibers of the diaphragm arise radially, from the margins of the inferior thoracic aperture, and converge into a large central tend.

Diaphragm

- Because of the oblique angle of the inferior thoracic aperture, the posterior attachment of the diaphragm is inferior to the anterior attachment.
- The diaphragm is not flat; rather, it 'balloons' superiorly, on both the right and left sides, to form domes.
- The right dome is higher than the left, reaching as far as rib V.

Diaphragm

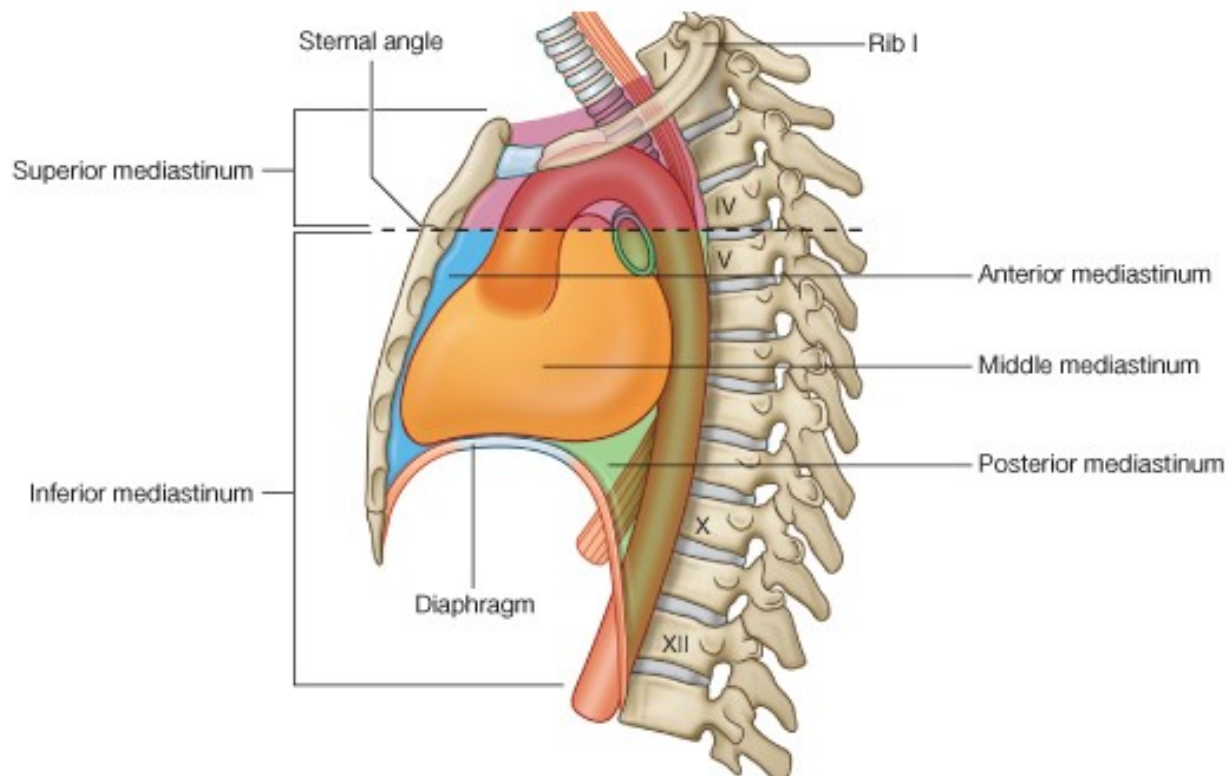
- As the diaphragm contracts, the height of the domes decreases and the volume of the thorax increases.
- The esophagus and inferior vena cava penetrate the diaphragm; the aorta passes posterior to the diaphragm.

Mediastinum

- The **mediastinum** is a thick midline partition that extends from the sternum anteriorly to the thoracic vertebrae posteriorly, and from the superior thoracic aperture to the inferior thoracic aperture.
- A horizontal plane passing through the sternal angle and the intervertebral disc between vertebrae TIV and TV separates the mediastinum into superior and inferior parts.
- The inferior part is further subdivided by the pericardium, which encloses the pericardial cavity surrounding the heart.
- The pericardium and heart constitute the middle mediastinum.

Pleural cavities

- The two pleural cavities are situated on either side of the mediastinum.



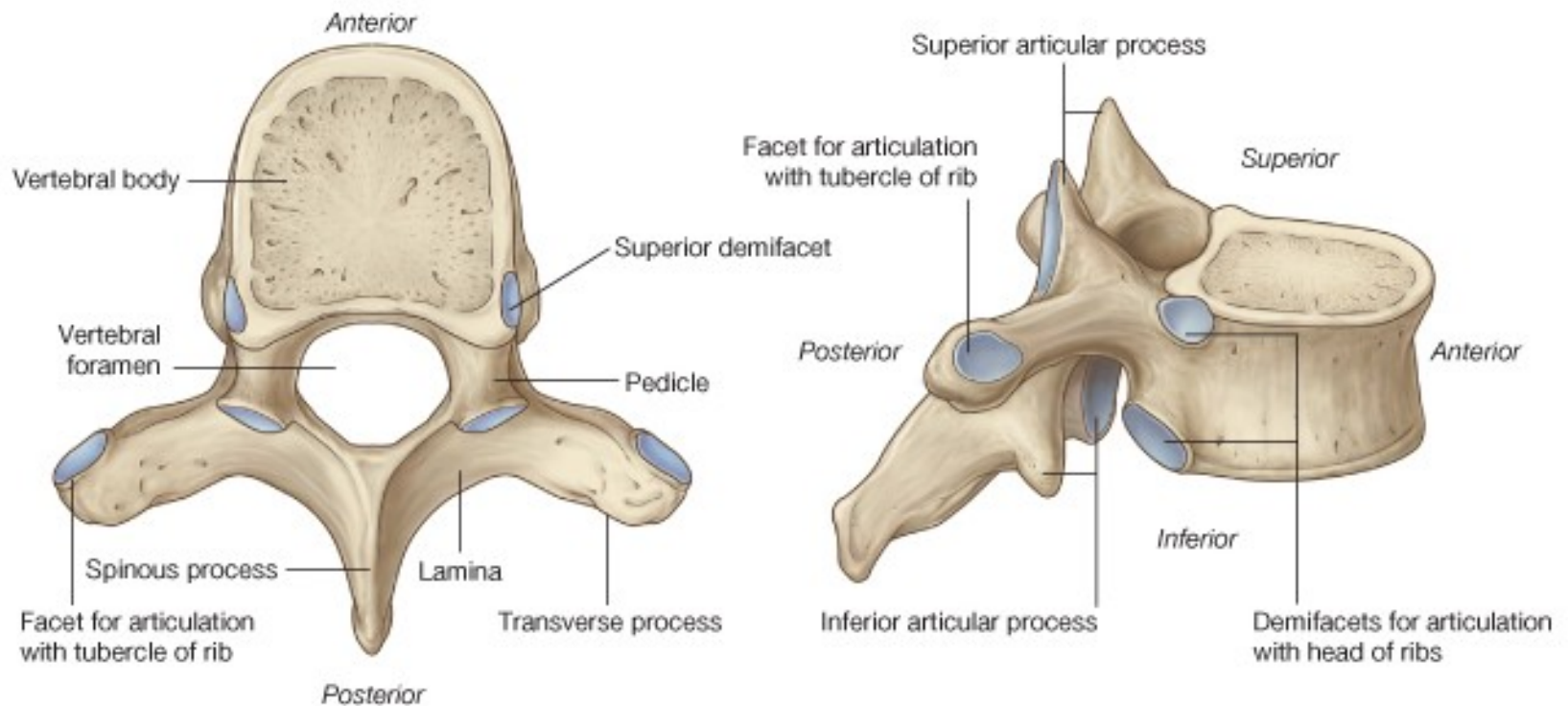
Pleural cavities

- Each **pleural cavity** is completely lined by a mesothelial membrane called the pleura.
- During development, the lungs grow out of the mediastinum, becoming surrounded by the pleural cavities.
- As a result, the outer surface of each organ is covered by pleura.

THORACIC WALL

- The thoracic wall is segmental in design and composed of skeletal elements and muscles. It extends between:
 - the superior thoracic aperture bordered by vertebra T1, rib I, and the manubrium of sternum; and
 - the inferior thoracic aperture bordered by vertebra TXII, rib XII, the end of rib XI, the costal margin, and the xiphoid process of sternum.

Typical thoracic vertebra



Skeletal framework

- The skeletal elements of the thoracic wall consist of thoracic vertebrae, intervertebral discs, ribs, and sternum.
- Thoracic vertebrae
- There are twelve **thoracic vertebrae**, each of which is characterized by articulations with ribs.

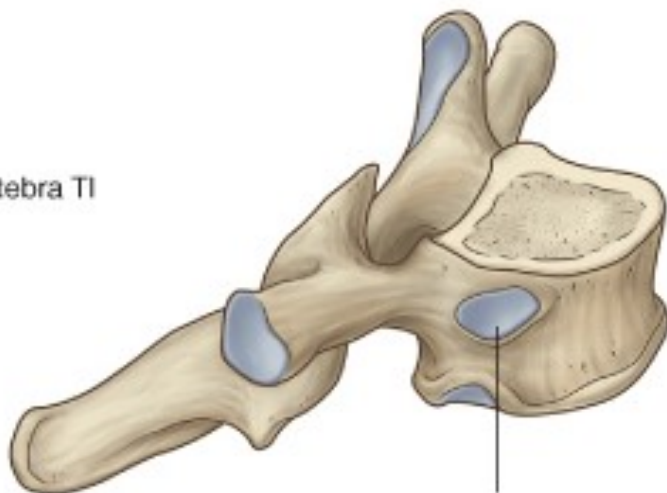
A typical thoracic vertebra

- A typical thoracic vertebra has a heart-shaped **vertebral body**, with roughly equal dimensions in the transverse and anteroposterior directions, and a long spinous process.
- The **vertebral foramen** is generally circular and the **laminae** are broad and overlap with those of the vertebra below.
- The **superior articular processes** are flat, with their articular surfaces facing almost directly posteriorly, while the **inferior articular processes** project from the laminae and their articular facets face anteriorly.
- The **transverse processes** are club shaped and project posterolaterally.

Articulation with ribs

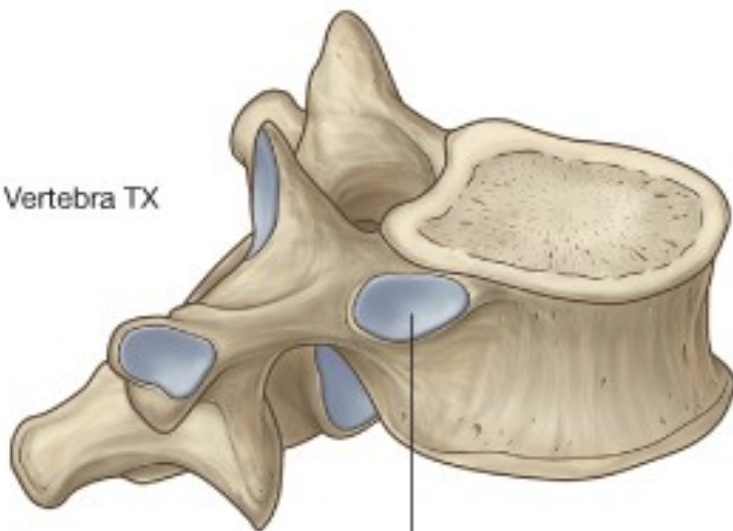
- A typical thoracic vertebra has three sites on each side for articulation with ribs:
 - two demifacets (i.e. partial facets) are located on the superior and inferior aspects of the body for articulation with corresponding sites on the heads of adjacent ribs.
 - The **superior costal facet** articulates with part of the head of its own rib, and the **inferior costal facet** articulates with part of the head of the rib below;
 - an oval facet (**transverse costal facet**) at the end of the transverse process articulates with the tubercle of its own rib.

Vertebra T1



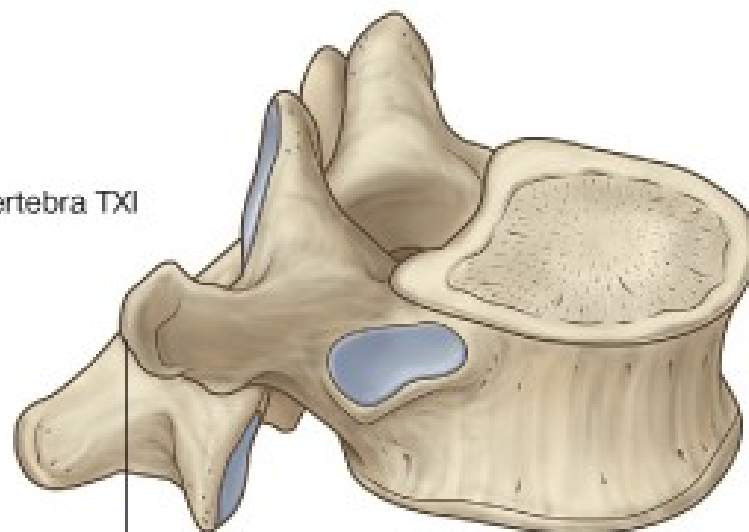
Superior costal facet for head of rib 1

Vertebra TX



Single complete costal facet for head of rib X

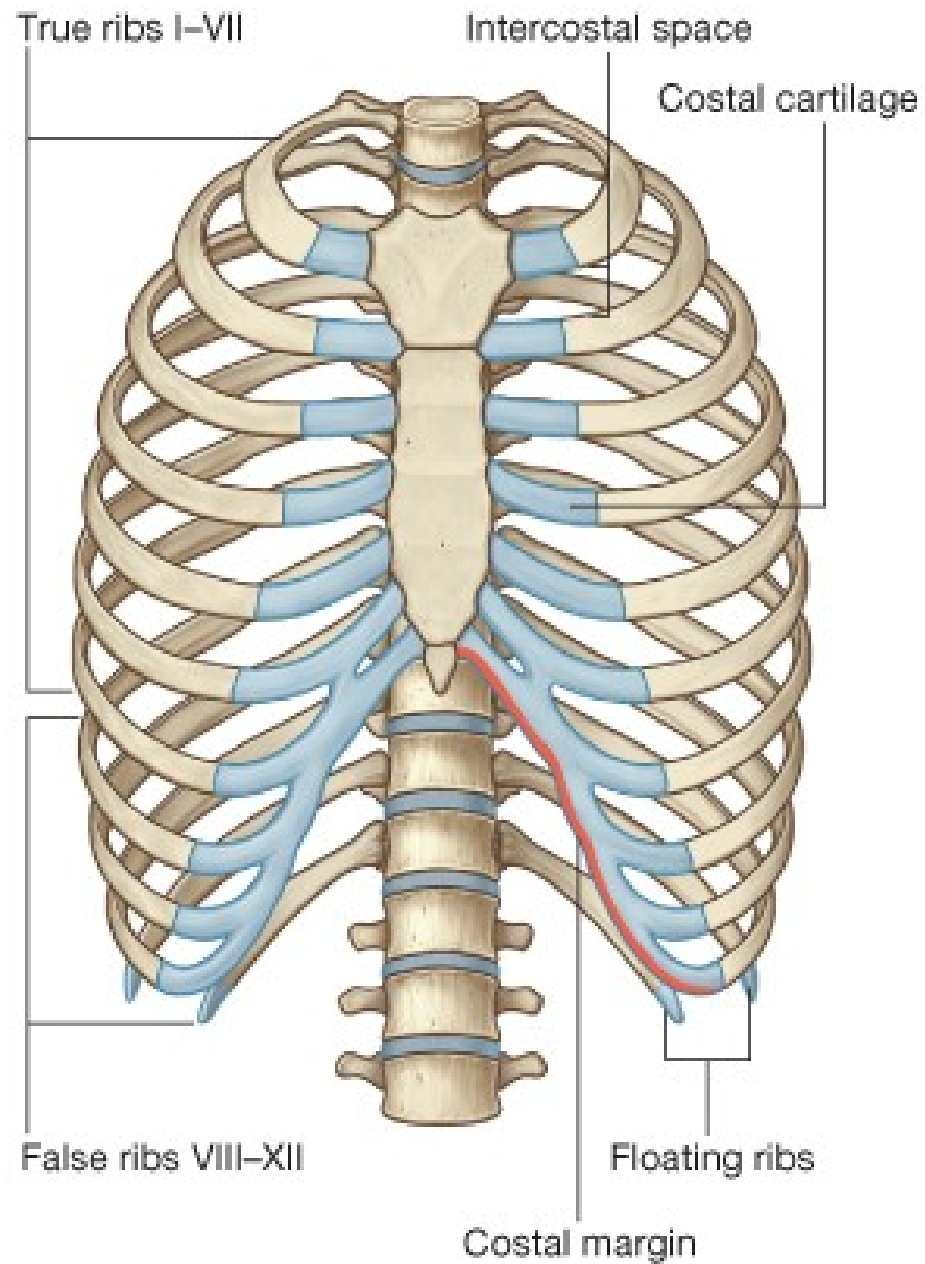
Vertebra TXI



No costal facet on transverse process

Ribs

- There are twelve pairs of ribs, each terminating anteriorly in a costal cartilage.
- Although all ribs articulate with the vertebral column, only the costal cartilages of the upper seven ribs, known as **true ribs**, articulate directly with the sternum.
- The remaining five pairs of ribs are **false ribs**:
 - Although all ribs articulate with the vertebral column, only the costal cartilages of the upper seven ribs, known as **true ribs**, articulate directly with the sternum.
 - The remaining five pairs of ribs are **false ribs**:



Ribs

- A typical rib consists of a curved shaft with anterior and posterior ends.
- The anterior end is continuous with its costal cartilage.
- The posterior end articulates with the vertebral column and is characterized by a head, neck, and tubercle.

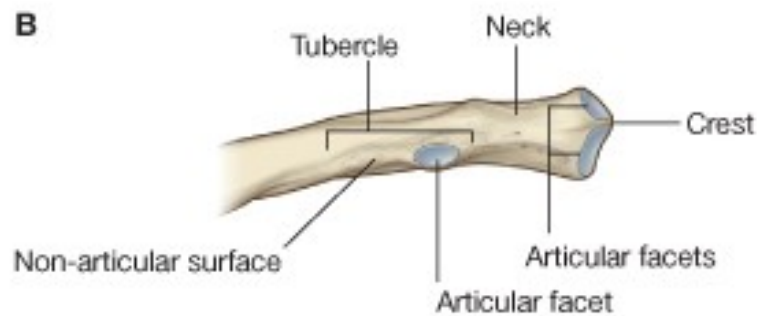
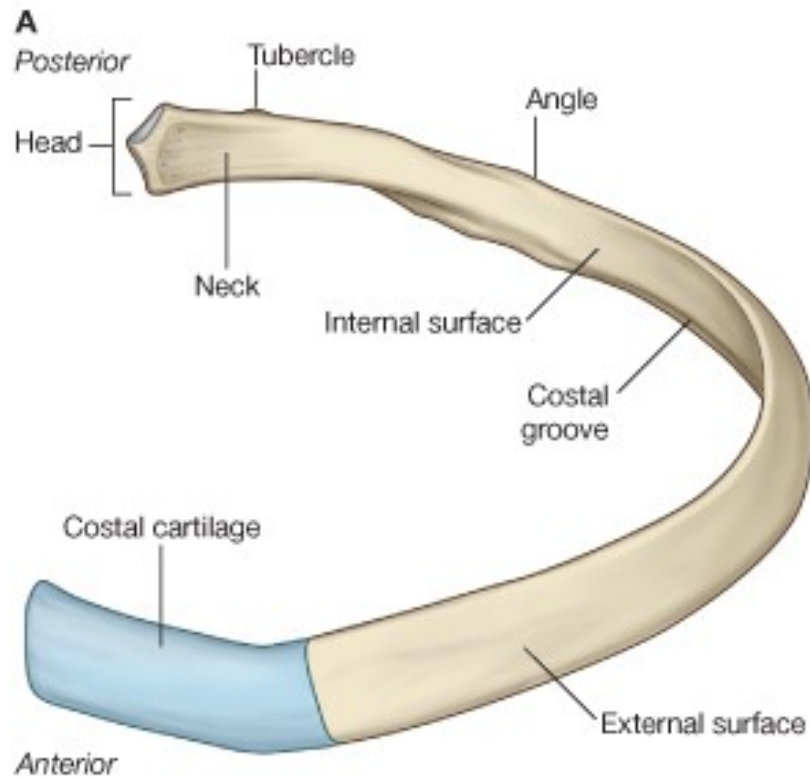
Ribs

- The **head** is somewhat expanded and typically presents two articular surfaces separated by a **crest**.
- The smaller superior surface articulates with the inferior costal facet on the body of the vertebra above, while the larger inferior facet articulates with the superior costal facet of its own vertebra.

Ribs

- The **neck** is a short flat region of bone that separates the head from the tubercle.
- The **tubercle** projects posteriorly from the junction of the neck with the shaft and consists of two regions, an articular part and a non-articular part:
 - the articular part is medial and has an oval facet for articulation with a corresponding facet on the transverse process of the associated vertebra;
 - the raised non-articular part is roughened by ligament attachments.

Ribs



Distinct features of upper and lower ribs

- The upper and lower ribs have distinct features.
- **Rib I** is flat in the horizontal plane and has broad superior and inferior surfaces.
- From its articulation with vertebra T1, it slopes inferiorly to its attachment to the manubrium of sternum..

Distinct features of upper and lower ribs

- The head articulates only with the body of vertebra T1 and therefore has only one articular surface.
- Like other ribs, the tubercle has a facet for articulation with the transverse process.
- The superior surface of the rib is characterized by a distinct tubercle, the **scalene tubercle**, which separates two smooth grooves that cross the rib approximately midway along the shaft.
- The anterior groove is caused by the subclavian vein, and the posterior groove is caused by the subclavian artery.
- Anterior and posterior to these grooves, the shaft is roughened by muscle and ligament attachments

Distinct features of upper and lower ribs

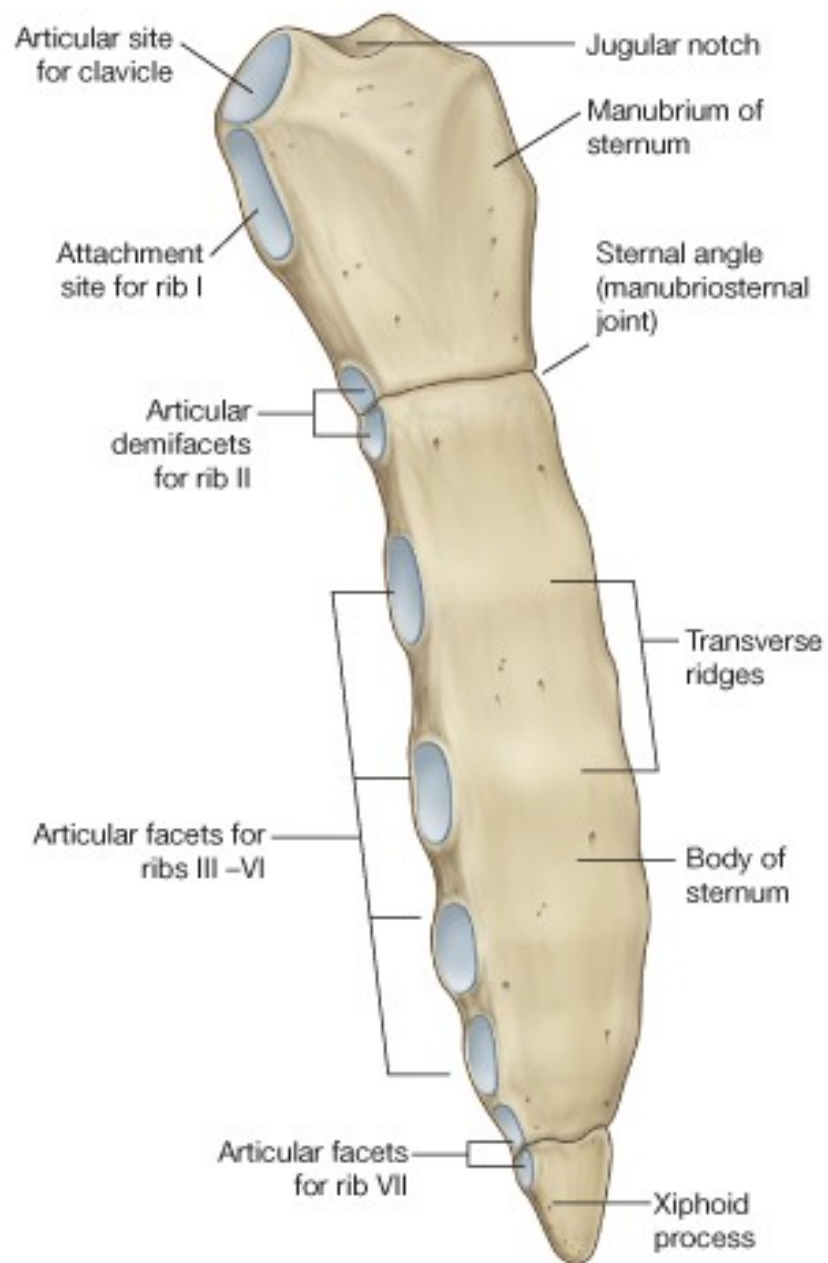
- **Rib II**, like rib I, is flat but twice as long. It articulates with the vertebral column in a way typical of most ribs.
- The head of rib X has a single facet for articulation with its own vertebra.
- **Ribs XI and XII** articulate only with the bodies of their own vertebrae and have no tubercles or necks.
- Both ribs are short, have little curve, and are pointed anteriorly.

Sternum

- The adult **sternum** consists of three major elements: the broad and superiorly positioned manubrium of sternum, the narrow and longitudinally oriented body of sternum, and the small and inferiorly positioned xiphoid process.

Manubrium of sternum

- The **manubrium of sternum** forms part of the bony framework of the neck and the thorax.
- The superior surface of the manubrium is expanded laterally and bears a distinct and palpable notch, the **jugular notch (suprasternal notch)**, in the midline.
- On either side of this notch is a large oval fossa for articulation with the clavicle.
- Immediately inferior to this fossa, on each lateral surface of the manubrium, is a facet for the attachment of the first costal cartilage.
- At the lower end of the lateral border is a demifacet for articulation with the upper half of the anterior end of the second costal cartilage.



Joints

- Costovertebral joints
- A typical rib articulates with:
 - the bodies of adjacent vertebrae, forming a joint with the head of the rib;
 - the transverse process of its related vertebra, forming a **costotransverse joint**

Joints

- Together, the costovertebral joints and related ligaments allow the necks of the ribs either to rotate around their longitudinal axes, which occurs mainly in the upper ribs, or to ascend and descend relative to the vertebral column, which occurs mainly in the lower ribs.
- The combined movements of all of the ribs on the vertebral column are essential for altering the volume of the thoracic cavity during breathing.

